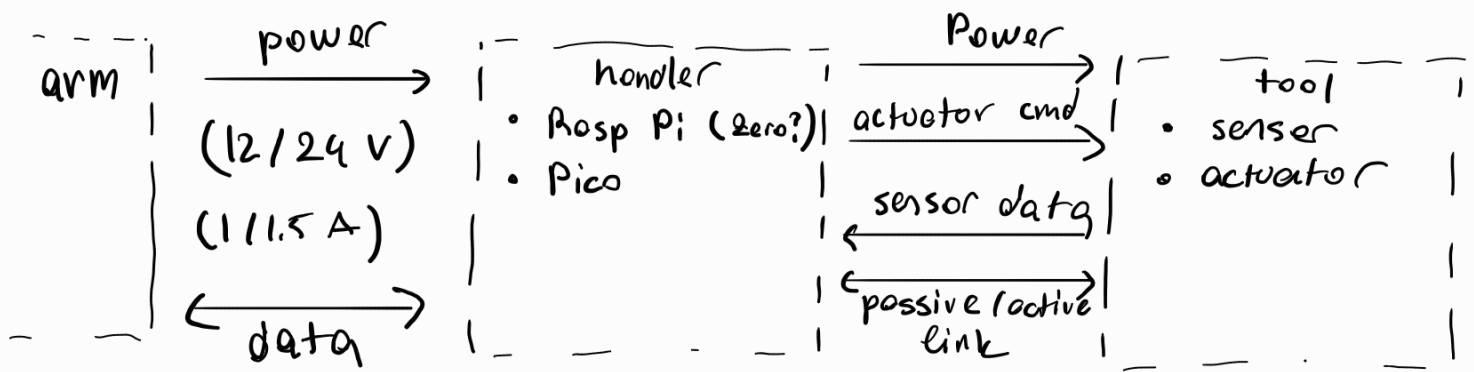


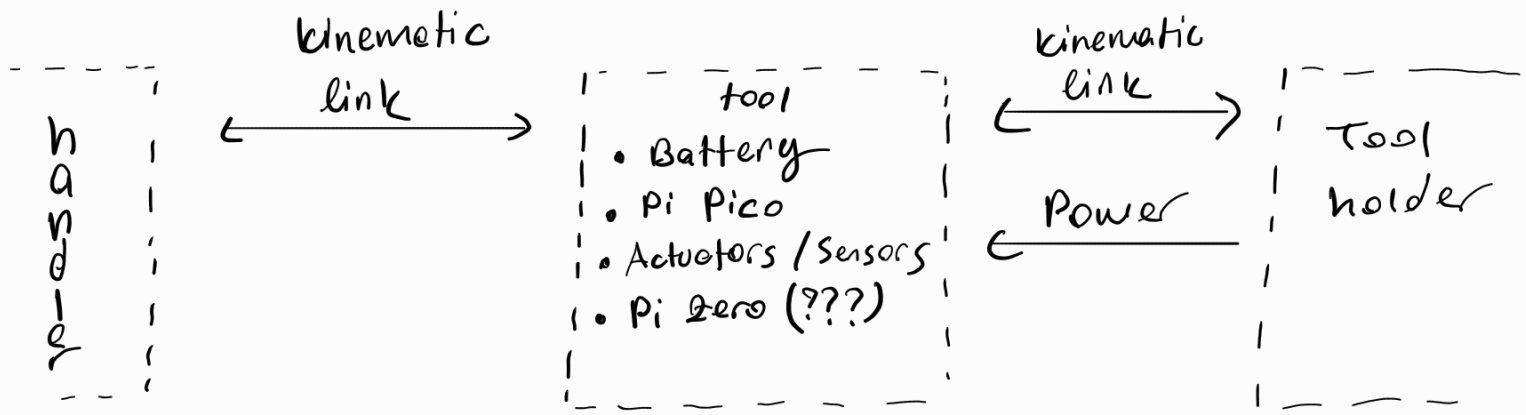
PCB

option 1 - take power from arm



- no need for battery management, basic PCB design
- plenty of power
- easy ROS integration and image processing with Pi
- transmitting power/data from handler to tool may be challenging
- design will be specific to UR5e
- Serial communication $\Rightarrow$ reliable
-

option 2 - battery powered



- Pi Zero for every tool for ROS (seems expensive)
- Use MicroROS with Pico instead? (may be challenging)
(or ESP32)
- Don't use ROS?
- 2 / 3 S battery at least (needs balancing?)
- Wireless communication may be problematic
- Tools may function independently from arm
- Recharging batteries with tool holder may be challenging
- Design will be less specific to UR5e

